

**Министерство науки и высшего образования Российской Федерации
ФЕДЕРАЛЬНОЕ ГОСУДАРСТВЕННОЕ АВТОНОМНОЕ ОБРАЗОВАТЕЛЬНОЕ
УЧРЕЖДЕНИЕ ВЫСШЕГО ОБРАЗОВАНИЯ
НАЦИОНАЛЬНЫЙ ИССЛЕДОВАТЕЛЬСКИЙ УНИВЕРСИТЕТ ИТМО**

ОТЧЕТ

ПО ЛАБОРАТОРНЫМ РАБОТАМ № 3.1, № 3.2

«Проектирование и реализация баз данных в СУБД PostgreSQL»

по дисциплине «Проектирование и реализация баз данных»

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Группа К3239

Направление подготовки 09.03.03 Прикладная информатика

Образовательная программа Мобильные и сетевые технологии 2024

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Цель работы: овладеть практическими навыками создания таблиц базы данных PostgreSQL 1X, заполнения их рабочими данными, резервного копирования и восстановления БД.

Практическое задание:

1. Установить СУБД PostgreSQL 1X.
2. Создать базу данных с использованием pgAdmin 4
3. Создать схему в составе базы данных.
4. Создать таблицы базы данных.
5. Установить ограничения на данные: Primary Key, Unique, Check, Foreign Key.
6. Заполнить таблицы БД рабочими данными.
7. Создать резервную копию БД. Указание:

Создать две резервные копии:

- с расширением CUSTOM для восстановления БД;
- с расширением для листинга (в отчете);
- при создании резервных копий БД настроить параметры Dump options для Type of objects и Queries.

8. Восстановить БД.

Выполнение:

БД “Аэропорт” (вариант 8)

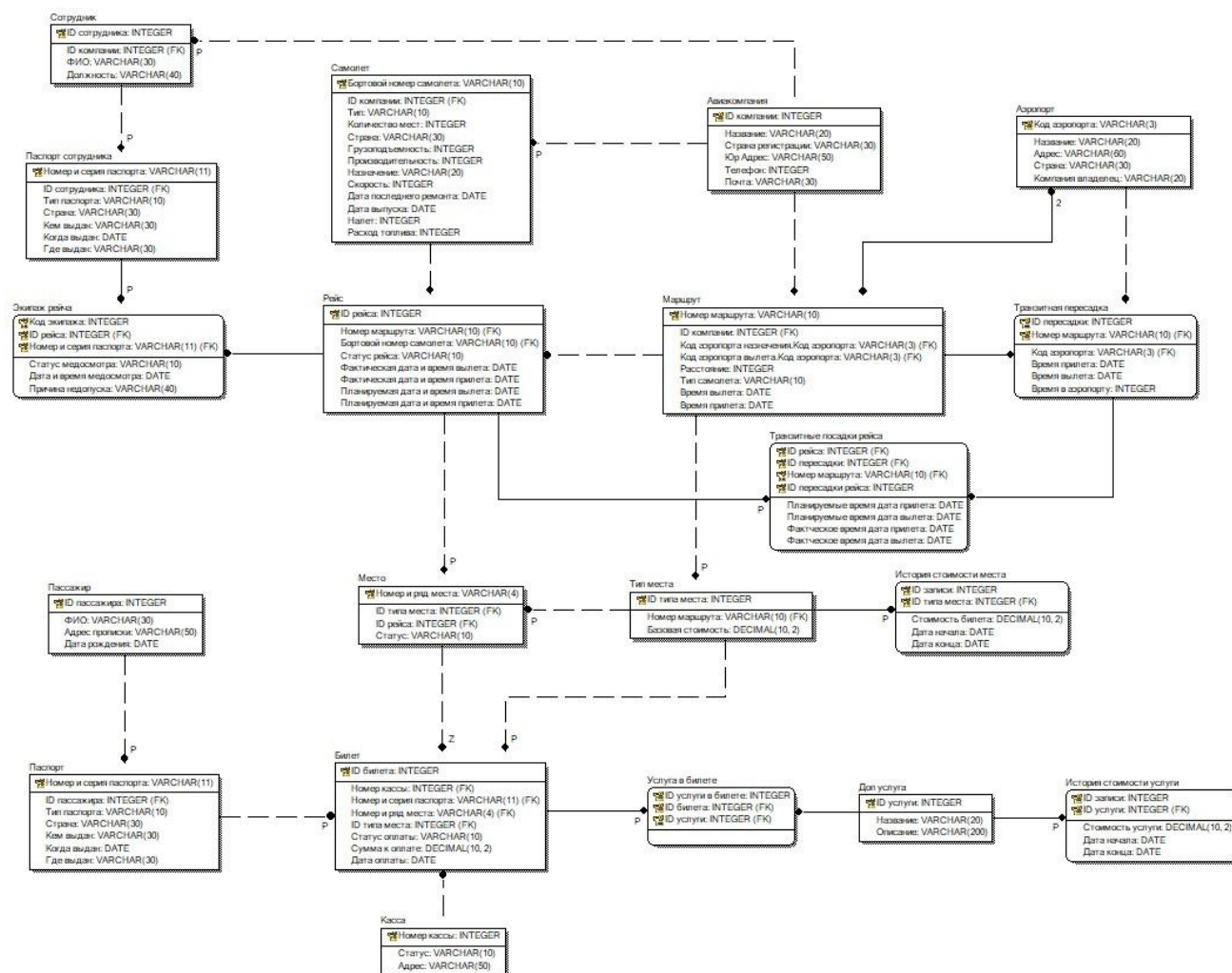


Рис. 1: Схема инфологической модели БД ЛР 2

На основе разработанной инфологической модели была построена физическая структура базы данных, с использованием встроенного функционала pgadmin 4. В соответствии с методическими указаниями составные ключи были заменены на простые суррогатные.

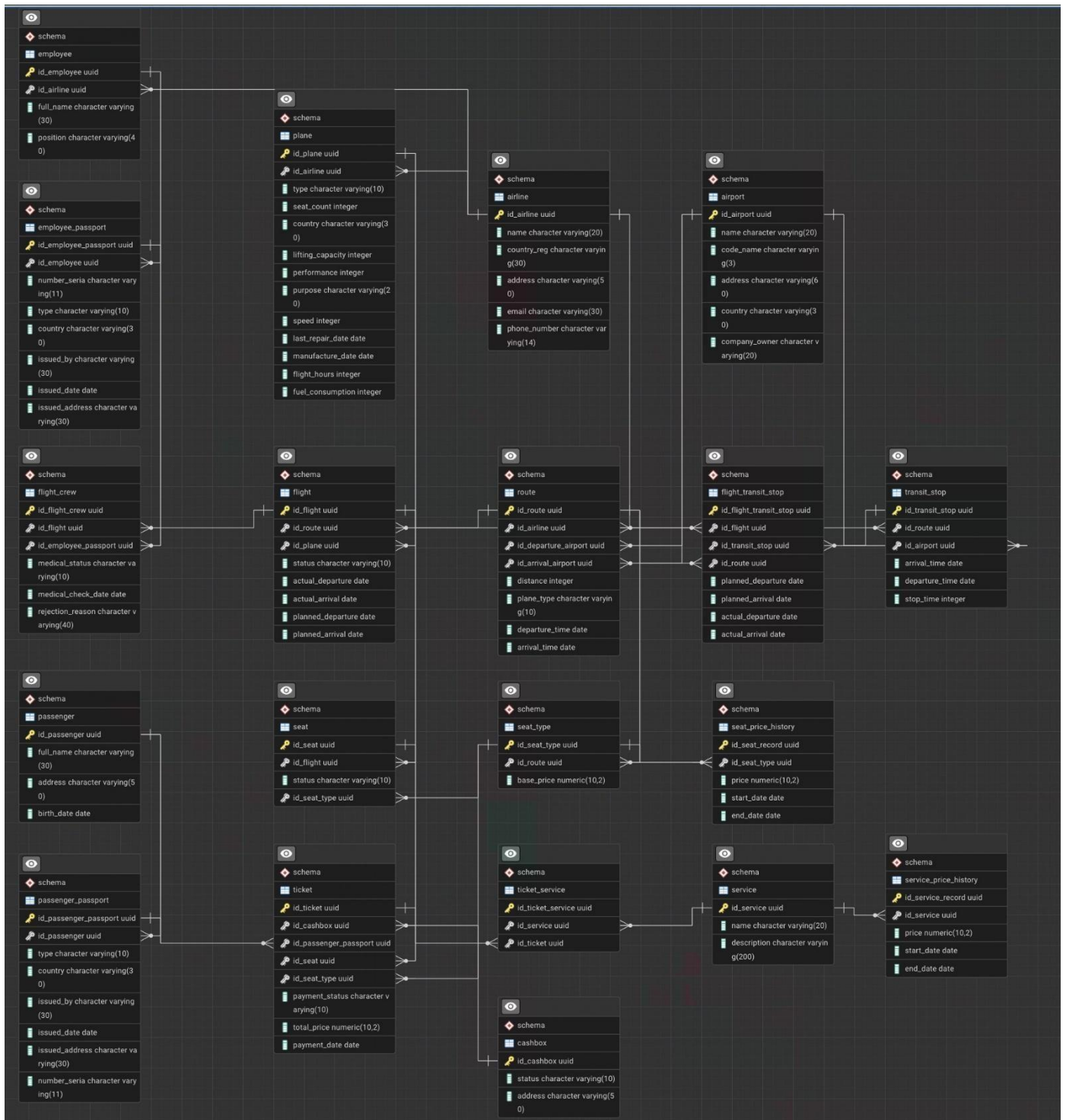


Рис. 2: ERD-диаграмма БД «Аэропорт»

Так же были использованы ограничения на данные:

- Ограничения на не пустое значение (NOT NULL)
- Ограничения на первичные и внешние ключи
- Ограничения на валидацию форматов (CHECK)

Далее таблицы были заполнены тестовыми данными с использованием SQL-запросов INSERT

	id_airport [PK] uuid	name character varying (20)	code_name character varying (3)	address character varying (60)	country character varying (30)	company_owner character varying (20)
186	d0e/UabU-0241-4a60-9cb0-e33cd6/e8...	Аэропорт 84	ADG	Адрес аэропорта 84	Россия	Владелец
187	61899d74-15c0-4731-a859-85dd78b78...	Аэропорт 85	ADH	Адрес аэропорта 85	США	Владелец
188	61f34a79-6260-4636-b2f8-c4b5fbf386...	Аэропорт 86	ADI	Адрес аэропорта 86	Европа	Владелец
189	781824dd-1c7b-499e-b1cd-ee51939dd...	Аэропорт 87	ADJ	Адрес аэропорта 87	Европа	Владелец
190	9f105cbb-e2e9-485f-8fec-d81c2c9d2d...	Аэропорт 88	ADK	Адрес аэропорта 88	Россия	Владелец
191	c94a0218-6a33-4a57-a49d-ac76bb3aa...	Аэропорт 89	ADL	Адрес аэропорта 89	США	Владелец
192	4c4782a9-c2eb-4662-96b1-59050a94d...	Аэропорт 90	ADM	Адрес аэропорта 90	Европа	Владелец
193	5adf63c9-b740-4e58-878e-f2e4a8f704...	Аэропорт 91	ADN	Адрес аэропорта 91	Европа	Владелец
194	a24ed389-998a-42a5-a162-92bc8827e...	Аэропорт 92	ADO	Адрес аэропорта 92	Россия	Владелец
195	b41e0012-92d8-4b17-9d3f-944e8332b...	Аэропорт 93	ADP	Адрес аэропорта 93	США	Владелец
196	3f0f04d4-eec3-41ed-a05c-6d2d238431...	Аэропорт 94	ADQ	Адрес аэропорта 94	Европа	Владелец
197	dbc0478e-045b-4114-bad5-00af9d030...	Аэропорт 95	ADR	Адрес аэропорта 95	Европа	Владелец
198	4d97f784-5c76-4f34-99cf-028d3f7825...	Аэропорт 96	ADS	Адрес аэропорта 96	Россия	Владелец
199	8b7c59f1-c012-4f42-a30f-329aa59f1f74	Аэропорт 97	ADT	Адрес аэропорта 97	США	Владелец
200	c05a8f21-3f6b-4899-bb14-df6c88a878...	Аэропорт 98	ADU	Адрес аэропорта 98	Европа	Владелец
201	0090c8c1-6301-4fc6-bb21-23ca16014...	Аэропорт 99	ADV	Адрес аэропорта 99	Европа	Владелец
202	3e94076d-879d-4574-b5cc-e978fcfa4b...	Аэропорт 100	ADW	Адрес аэропорта 100	Россия	Владелец
Total rows: 202		Query complete 00:00:00.098				

Рис. 3: Наполнение таблицы аэропортов

В соответствии с требованиями ЛР №3.2 были созданы два дампа:

- airport_lab3_custom.backup — для восстановления через pgAdmin.
- airport_lab3_plain.sql — для анализа SQL-кода.

Содержимое airport_lab3_plain.sql:

--

-- PostgreSQL database dump

--

\restrict

wM96tMIVekC04YxgAZzNHuX3LhcoxbsVtJ7S4wQFoFBZOvFPTQmOTYDq7
CPa0W5

-- Dumped from database version 14.22

-- Dumped by pg_dump version 14.22

-- Started on 2026-03-30 13:03:29 MSK

SET statement_timeout = 0;

SET lock_timeout = 0;

SET idle_in_transaction_session_timeout = 0;

```
SET client_encoding = 'UTF8';
SET standard_conforming_strings = on;
```

```
SELECT pg_catalog.set_config('search_path', '', false);
SET check_function_bodies = false;
SET xmloption = content;
SET client_min_messages = warning;
SET row_security = off;
```

```
--
-- TOC entry 4 (class 2615 OID 16554)
-- Name: schema; Type: SCHEMA; Schema: -; Owner: postgres
--
```

```
CREATE SCHEMA schema;
```

```
ALTER SCHEMA schema OWNER TO postgres;
```

```
SET default_tablespace = '';
```

```
SET default_table_access_method = heap;
```

```
--
-- TOC entry 210 (class 1259 OID 16560)
-- Name: airline; Type: TABLE; Schema: schema; Owner: postgres
--
```

```
CREATE TABLE schema.airline (
    id_airline uuid NOT NULL,
    name character varying(20) NOT NULL,
    country_reg character varying(30) NOT NULL,
    address character varying(50) NOT NULL,
    email character varying(30) NOT NULL,
    phone_number character varying(14)
);
```

```
ALTER TABLE schema.airline OWNER TO postgres;
```

```
--  
-- TOC entry 214 (class 1259 OID 16595)  
-- Name: airport; Type: TABLE; Schema: schema; Owner: postgres  
--
```

```
CREATE TABLE schema.airport (  
    id_airport uuid NOT NULL,  
    name character varying(20) NOT NULL,  
    code_name character varying(3) NOT NULL,  
    address character varying(60) NOT NULL,  
    country character varying(30) NOT NULL,  
    company_owner character varying(20) NOT NULL  
);
```

```
ALTER TABLE schema.airport OWNER TO postgres;
```

```
--  
-- TOC entry 225 (class 1259 OID 16762)  
-- Name: cashbox; Type: TABLE; Schema: schema; Owner: postgres  
--
```

```
CREATE TABLE schema.cashbox (  
    id_cashbox uuid NOT NULL,  
    status character varying(10) NOT NULL,  
    address character varying(50) NOT NULL  
);
```

```
ALTER TABLE schema.cashbox OWNER TO postgres;
```

```
--  
-- TOC entry 211 (class 1259 OID 16570)  
-- Name: employee; Type: TABLE; Schema: schema; Owner: postgres  
--
```

```
CREATE TABLE schema.employee (  
    id_employee uuid NOT NULL,
```

```
id_airline uuid NOT NULL,  
full_name character varying(30) NOT NULL,  
"position" character varying(40) NOT NULL  
);
```

```
ALTER TABLE schema.employee OWNER TO postgres;
```

```
--  
-- TOC entry 212 (class 1259 OID 16580)  
-- Name: employee_passport; Type: TABLE; Schema: schema; Owner: postgres  
--
```

```
CREATE TABLE schema.employee_passport (  
    id_employee_passport uuid NOT NULL,  
    id_employee uuid NOT NULL,  
    number_seria character varying(11) NOT NULL,  
    type character varying(10) NOT NULL,  
    country character varying(30) NOT NULL,  
    issued_by character varying(30) NOT NULL,  
    issued_date date NOT NULL,  
    issued_address character varying(30) NOT NULL  
);
```

```
ALTER TABLE schema.employee_passport OWNER TO postgres;
```

```
--  
-- TOC entry 216 (class 1259 OID 16640)  
-- Name: flight; Type: TABLE; Schema: schema; Owner: postgres  
--
```

```
CREATE TABLE schema.flight (  
    id_flight uuid NOT NULL,  
    id_route uuid NOT NULL,  
    id_plane uuid NOT NULL,  
    status character varying(10) NOT NULL,  
    actual_departure date NOT NULL,
```

```
    actual_arrival date NOT NULL,  
    planned_departure date NOT NULL,  
    planned_arrival date NOT NULL  
);
```

```
ALTER TABLE schema.flight OWNER TO postgres;
```

```
--  
-- TOC entry 213 (class 1259 OID 16590)  
-- Name: flight_crew; Type: TABLE; Schema: schema; Owner: postgres
```

```
--
```

```
CREATE TABLE schema.flight_crew (  
    id_flight_crew uuid NOT NULL,  
    id_flight uuid NOT NULL,  
    id_employee_passport uuid NOT NULL,  
    medical_status character varying(10) NOT NULL,  
    medical_check_date date NOT NULL,  
    rejection_reason character varying(40)  
);
```

```
ALTER TABLE schema.flight_crew OWNER TO postgres;
```

```
--  
-- TOC entry 218 (class 1259 OID 16680)  
-- Name: flight_transit_stop; Type: TABLE; Schema: schema; Owner: postgres  
--
```

```
CREATE TABLE schema.flight_transit_stop (  
    id_flight_transit_stop uuid NOT NULL,  
    id_flight uuid NOT NULL,  
    id_transit_stop uuid NOT NULL,  
    id_route uuid NOT NULL,  
    planned_departure date NOT NULL,  
    planned_arrival date NOT NULL,  
    actual_departure date,
```

```
    actual_arrival date
);
```

```
ALTER TABLE schema.flight_transit_stop OWNER TO postgres;
```

```
--
-- TOC entry 219 (class 1259 OID 16700)
-- Name: passenger; Type: TABLE; Schema: schema; Owner: postgres
--
```

```
CREATE TABLE schema.passenger (
    id_passenger uuid NOT NULL,
    full_name character varying(30) NOT NULL,
```

```
    address character varying(50) NOT NULL,
    birth_date date NOT NULL
);
```

```
ALTER TABLE schema.passenger OWNER TO postgres;
```

```
--
-- TOC entry 220 (class 1259 OID 16705)
-- Name: passenger_passport; Type: TABLE; Schema: schema; Owner: postgres
--
```

```
CREATE TABLE schema.passenger_passport (
    id_passenger_passport uuid NOT NULL,
    id_passenger uuid NOT NULL,
    type character varying(10) NOT NULL,
    country character varying(30) NOT NULL,
    issued_by character varying(30) NOT NULL,
    issued_date date NOT NULL,
    issued_address character varying(30) NOT NULL,
    number_seria character varying(11) NOT NULL
);
```

```
ALTER TABLE schema.passenger_passport OWNER TO postgres;
```

```
--
```

```
-- TOC entry 209 (class 1259 OID 16555)
```

```
-- Name: plane; Type: TABLE; Schema: schema; Owner: postgres
```

```
--
```

```
CREATE TABLE schema.plane (  
    id_plane uuid NOT NULL,  
    id_airline uuid NOT NULL,  
    type character varying(10) NOT NULL,  
    seat_count integer NOT NULL,  
    country character varying(30) NOT NULL,  
    lifting_capacity integer NOT NULL,  
    performance integer NOT NULL,  
    purpose character varying(20) NOT NULL,  
    speed integer NOT NULL,
```

```
    last_repair_date date,  
    manufacture_date date NOT NULL,  
    flight_hours integer NOT NULL,  
    fuel_consumption integer NOT NULL  
);
```

```
ALTER TABLE schema.plane OWNER TO postgres;
```

```
--
```

```
-- TOC entry 215 (class 1259 OID 16620)
```

```
-- Name: route; Type: TABLE; Schema: schema; Owner: postgres
```

```
--
```

```
CREATE TABLE schema.route (  
    id_route uuid NOT NULL,  
    id_airline uuid NOT NULL,  
    id_departure_airport uuid NOT NULL,  
    id_arrival_airport uuid NOT NULL,  
    distance integer NOT NULL,  
    plane_type character varying(10) NOT NULL,
```

```
    departure_time date NOT NULL,  
    arrival_time date NOT NULL  
);
```

```
ALTER TABLE schema.route OWNER TO postgres;
```

```
--  
-- TOC entry 221 (class 1259 OID 16715)  
-- Name: seat; Type: TABLE; Schema: schema; Owner: postgres  
--
```

```
CREATE TABLE schema.seat (  
    id_seat uuid NOT NULL,  
    id_flight uuid NOT NULL,  
    status character varying(10) NOT NULL,  
    id_seat_type uuid NOT NULL  
);
```

```
ALTER TABLE schema.seat OWNER TO postgres;
```

```
--  
-- TOC entry 223 (class 1259 OID 16747)  
-- Name: seat_price_history; Type: TABLE; Schema: schema; Owner: postgres  
--
```

```
CREATE TABLE schema.seat_price_history (  
    id_seat_record uuid NOT NULL,  
    id_seat_type uuid NOT NULL,  
    price numeric(10,2) NOT NULL,  
    start_date date NOT NULL,  
    end_date date NOT NULL  
);
```

```
ALTER TABLE schema.seat_price_history OWNER TO postgres;
```

```
--  
-- TOC entry 222 (class 1259 OID 16720)  
-- Name: seat_type; Type: TABLE; Schema: schema; Owner: postgres  
--
```

```
CREATE TABLE schema.seat_type (  
    id_seat_type uuid NOT NULL,  
    id_route uuid NOT NULL,  
    base_price numeric(10,2) NOT NULL  
);
```

```
ALTER TABLE schema.seat_type OWNER TO postgres;
```

```
--  
-- TOC entry 226 (class 1259 OID 16787)  
-- Name: service; Type: TABLE; Schema: schema; Owner: postgres  
--
```

```
CREATE TABLE schema.service (  
    id_service uuid NOT NULL,  
    name character varying(20) NOT NULL,  
    description character varying(200) NOT NULL  
  
);
```

```
ALTER TABLE schema.service OWNER TO postgres;
```

```
--  
-- TOC entry 228 (class 1259 OID 16810)  
-- Name: service_price_history; Type: TABLE; Schema: schema; Owner: postgres  
--
```

```
CREATE TABLE schema.service_price_history (  
    id_service_record uuid NOT NULL,  
    id_service uuid NOT NULL,  
    price numeric(10,2) NOT NULL,  
    start_date date NOT NULL,
```

```
    end_date date NOT NULL
);
```

```
ALTER TABLE schema.service_price_history OWNER TO postgres;
```

```
--
-- TOC entry 224 (class 1259 OID 16757)
-- Name: ticket; Type: TABLE; Schema: schema; Owner: postgres
--
```

```
CREATE TABLE schema.ticket (
    id_ticket uuid NOT NULL,
    id_cashbox uuid NOT NULL,
    id_passenger_passport uuid NOT NULL,
    id_seat uuid NOT NULL,
    id_seat_type uuid NOT NULL,
    payment_status character varying(10) NOT NULL,
    total_price numeric(10,2) NOT NULL,
    payment_date date NOT NULL
);
```

```
ALTER TABLE schema.ticket OWNER TO postgres;
```

```
--
--
-- TOC entry 227 (class 1259 OID 16792)
-- Name: ticket_service; Type: TABLE; Schema: schema; Owner: postgres
--
```

```
CREATE TABLE schema.ticket_service (
    id_ticket_service uuid NOT NULL,
    id_service uuid NOT NULL,
    id_ticket uuid NOT NULL
);
```

```
ALTER TABLE schema.ticket_service OWNER TO postgres;
```

```
--
-- TOC entry 217 (class 1259 OID 16665)
-- Name: transit_stop; Type: TABLE; Schema: schema; Owner: postgres
--
```

```
CREATE TABLE schema.transit_stop (
    id_transit_stop uuid NOT NULL,
    id_route uuid NOT NULL,
    id_airport uuid NOT NULL,
    arrival_time date NOT NULL,
    departure_time date NOT NULL,
    stop_time integer NOT NULL
);
```

```
ALTER TABLE schema.transit_stop OWNER TO postgres;
```

```
--
-- TOC entry 3774 (class 0 OID 16560)
-- Dependencies: 210
-- Data for Name: airline; Type: TABLE DATA; Schema: schema; Owner:
postgres
--
```

```
COPY schema.airline (id_airline, name, country_reg, address, email,
phone_number) FROM stdin;
7b9c4945-82ab-4544-93e7-17fed61c9c25 Победа Россия Ул.
Пушкина, д. 23к2 email@mail.ru +799999999999
```

```
5364618a-04d1-45df-819c-67529239a73c Победа Россия Москва ул.
Пушкина, д. 23к2 email@mail.ru +799999999999
4604ccbe-3324-48e2-be4d-7b56fd065fc2 Победа Россия Москва, ул.
Пушкина, д. 23к2 email@mail.ru +799999999999
417d352a-c149-4106-9c2a-b9e97364b61b Победа Россия Москва, ул.
Пушкина, д. 23/2 email@mail.ru +799999999999
21777d1f-bc73-491f-854e-ab25dfd8119a Победа Россия Москва, ул.
Пушкина, д. 23/2 email@mail.ru +799999999999
```

886be670-cd06-4a70-86c5-5720bfee3fff	Победа	Россия	Москва, ул.
Пушкина, д. 23/2 email@mail.ru +799999999999			
9c1abe78-0799-401d-9670-8635c3d9931d	Победа	Россия	Москва, ул.
Пушкина, д. 23/2 email@mail.ru +799999999999			
223da672-4bac-4eca-813a-68fbd14112f8	Победа	Россия	Москва, ул.
Пушкина, д. 23/2 email@mail.ru \N			
f0e9da01-534f-4b18-a79b-4580c94c0ec2	Победа	Россия	Москва, ул.
Пушкина, д. 23/2 email@mail.ru \N			
6dda3482-df1f-4ef0-8b1b-9d9ef8742063	name coun	address	
sdf@adsf.rsdvf \N			
5ec05804-4997-4e89-afb3-712cc4694541	name coun	address	
sdf@adsf.rsdvf \N			
78ba690c-a979-4683-aa51-2d3626099ca0	name coun	address	
sdf@adsf.rsdvf \N			

\.

```
--
-- TOC entry 3778 (class 0 OID 16595)
-- Dependencies: 214
-- Data for Name: airport; Type: TABLE DATA; Schema: schema; Owner:
postgres
```

```
COPY schema.airport (id_airport, name, code_name, address, country,
company_owner) FROM stdin;
4665fe02-292a-4561-b283-c97cf7261903 name NMA address country
owner
bb876138-9607-4f52-b1cd-99f2a530b102 name NMA address country
owner
```

\.

```
--
-- TOC entry 3789 (class 0 OID 16762)

-- Dependencies: 225
-- Data for Name: cashbox; Type: TABLE DATA; Schema: schema; Owner:
postgres
--
```

```
COPY schema.cashbox (id_cashbox, status, address) FROM stdin;  
\.
```

```
--  
-- TOC entry 3775 (class 0 OID 16570)  
-- Dependencies: 211  
-- Data for Name: employee; Type: TABLE DATA; Schema: schema; Owner:  
postgres  
--
```

```
COPY schema.employee (id_employee, id_airline, full_name, "position") FROM  
stdin;  
\.
```

```
--  
-- TOC entry 3776 (class 0 OID 16580)  
-- Dependencies: 212  
-- Data for Name: employee_passport; Type: TABLE DATA; Schema: schema;  
Owner: postgres  
--
```

```
COPY schema.employee_passport (id_employee_passport, id_employee,  
number_seria, type, country, issued_by, issued_date, issued_address) FROM stdin;  
\.
```

```
--  
-- TOC entry 3780 (class 0 OID 16640)  
-- Dependencies: 216  
-- Data for Name: flight; Type: TABLE DATA; Schema: schema; Owner: postgres  
--
```

```
COPY schema.flight (id_flight, id_route, id_plane, status, actual_departure,  
actual_arrival, planned_departure, planned_arrival) FROM stdin;  
\.
```

```
--
-- TOC entry 3777 (class 0 OID 16590)
-- Dependencies: 213
-- Data for Name: flight_crew; Type: TABLE DATA; Schema: schema; Owner:
postgres
```

```
--
COPY schema.flight_crew (id_flight_crew, id_flight, id_employee_passport,
medical_status, medical_check_date, rejection_reason) FROM stdin;
\.
```

```
--
-- TOC entry 3782 (class 0 OID 16680)
-- Dependencies: 218
-- Data for Name: flight_transit_stop; Type: TABLE DATA; Schema: schema;
Owner: postgres
```

```
--
COPY schema.flight_transit_stop (id_flight_transit_stop, id_flight, id_transit_stop,
id_route, planned_departure, planned_arrival, actual_departure, actual_arrival)
FROM stdin;
\.
```

```
--
-- TOC entry 3783 (class 0 OID 16700)
-- Dependencies: 219
-- Data for Name: passenger; Type: TABLE DATA; Schema: schema; Owner:
postgres
```

```
--
COPY schema.passenger (id_passenger, full_name, address, birth_date) FROM
stdin;
b767047f-eb20-4017-bc24-fc75182a6d64   dfsd sdf      sdfsd2020-02-20
\.
```

```
--
-- TOC entry 3784 (class 0 OID 16705)
-- Dependencies: 220
```

-- Data for Name: passenger_passport; Type: TABLE DATA; Schema: schema;
Owner: postgres

--

COPY schema.passenger_passport (id_passenger_passport, id_passenger, type,
country, issued_by, issued_date, issued_address, number_seria) FROM stdin;
\\.

--

-- TOC entry 3773 (class 0 OID 16555)
-- Dependencies: 209
-- Data for Name: plane; Type: TABLE DATA; Schema: schema; Owner: postgres
--

COPY schema.plane (id_plane, id_airline, type, seat_count, country,
lifting_capacity, performance, purpose, speed, last_repair_date, manufacture_date,
flight_hours, fuel_consumption) FROM stdin;
\\.

--

-- TOC entry 3779 (class 0 OID 16620)
-- Dependencies: 215
-- Data for Name: route; Type: TABLE DATA; Schema: schema; Owner: postgres
--

COPY schema.route (id_route, id_airline, id_departure_airport, id_arrival_airport,
distance, plane_type, departure_time, arrival_time) FROM stdin;
\\.

--

-- TOC entry 3785 (class 0 OID 16715)
-- Dependencies: 221
-- Data for Name: seat; Type: TABLE DATA; Schema: schema; Owner: postgres
--

COPY schema.seat (id_seat, id_flight, status, id_seat_type) FROM stdin;

\.

--

-- TOC entry 3787 (class 0 OID 16747)

-- Dependencies: 223

-- Data for Name: seat_price_history; Type: TABLE DATA; Schema: schema;
Owner: postgres

--

COPY schema.seat_price_history (id_seat_record, id_seat_type, price, start_date,
end_date) FROM stdin;

\.

--

-- TOC entry 3786 (class 0 OID 16720)

-- Dependencies: 222

-- Data for Name: seat_type; Type: TABLE DATA; Schema: schema; Owner:
postgres

--

COPY schema.seat_type (id_seat_type, id_route, base_price) FROM stdin;

\.

--

-- TOC entry 3790 (class 0 OID 16787)

-- Dependencies: 226

-- Data for Name: service; Type: TABLE DATA; Schema: schema; Owner:
postgres

--

COPY schema.service (id_service, name, description) FROM stdin;

\.

--

```
-- TOC entry 3792 (class 0 OID 16810)
-- Dependencies: 228
-- Data for Name: service_price_history; Type: TABLE DATA; Schema: schema;
Owner: postgres
--
```

```
COPY schema.service_price_history (id_service_record, id_service, price,
start_date, end_date) FROM stdin;
```

```
\.
```

```
--
-- TOC entry 3788 (class 0 OID 16757)
-- Dependencies: 224
-- Data for Name: ticket; Type: TABLE DATA; Schema: schema; Owner: postgres
--
```

```
COPY schema.ticket (id_ticket, id_cashbox, id_passenger_passport, id_seat,
id_seat_type, payment_status, total_price, payment_date) FROM stdin;
\.
```

```
--
-- TOC entry 3791 (class 0 OID 16792)
-- Dependencies: 227
-- Data for Name: ticket_service; Type: TABLE DATA; Schema: schema; Owner:
postgres
--
```

```
COPY schema.ticket_service (id_ticket_service, id_service, id_ticket) FROM
stdin;
\.
```

```
--
-- TOC entry 3781 (class 0 OID 16665)
-- Dependencies: 217
```

-- Data for Name: transit_stop; Type: TABLE DATA; Schema: schema; Owner: postgres

--

COPY schema.transit_stop (id_transit_stop, id_route, id_airport, arrival_time, departure_time, stop_time) FROM stdin;

\\.

--

-- TOC entry 3578 (class 2606 OID 16599)

-- Name: airport airport_pkey; Type: CONSTRAINT; Schema: schema; Owner: postgres

--

ALTER TABLE ONLY schema.airport

ADD CONSTRAINT airport_pkey PRIMARY KEY (id_airport);

--

-- TOC entry 3522 (class 2606 OID 16870)

-- Name: employee_passport check_country; Type: CHECK CONSTRAINT; Schema: schema; Owner: postgres

--

ALTER TABLE schema.employee_passport

ADD CONSTRAINT check_country CHECK (((country)::text ~ '^[A-Яa-яA-Za-z .\\-]+\$',::text)) NOT VALID;

--

-- TOC entry 3538 (class 2606 OID 16876)

-- Name: flight check_actual_dates; Type: CHECK CONSTRAINT; Schema: schema; Owner: postgres

--

ALTER TABLE schema.flight

```
ADD CONSTRAINT check_actual_dates CHECK ((actual_departure <
actual_arrival)) NOT VALID;
```

```
--
```

```
-- TOC entry 3543 (class 2606 OID 16879)
```

```
-- Name: flight_transit_stop check_actual_dates; Type: CHECK CONSTRAINT;
Schema: schema; Owner: postgres
```

```
--
```

```
ALTER TABLE schema.flight_transit_stop
```

```
ADD CONSTRAINT check_actual_dates CHECK ((actual_departure <
actual_arrival)) NOT VALID;
```

```
--
```

```
-- TOC entry 3530 (class 2606 OID 16861)
```

```
-- Name: airport check_address; Type: CHECK CONSTRAINT; Schema: schema;
Owner: postgres
```

```
--
```

```
ALTER TABLE schema.airport
```

```
ADD CONSTRAINT check_address CHECK (((address)::text ~ '^[A-Яa-яA-
Za-z .,\-]+$',::text)) NOT VALID;
```

```
--
```

```
-- TOC entry 3561 (class 2606 OID 16865)
```

```
-- Name: cashbox check_address; Type: CHECK CONSTRAINT; Schema:
schema; Owner: postgres
```

```
--
```

```
ALTER TABLE schema.cashbox
```

```
ADD CONSTRAINT check_address CHECK (((address)::text ~ '^[A-Яa-яA-
Za-z .,\-]+$',::text)) NOT VALID;
```

```
--
```

```
-- TOC entry 3545 (class 2606 OID 16883)
```

```
-- Name: passenger check_address; Type: CHECK CONSTRAINT; Schema:
schema; Owner: postgres
```

```
--
```

```
ALTER TABLE schema.passenger
```

```
    ADD CONSTRAINT check_address CHECK (((address)::text ~ '^[A-Яa-яA-
Za-z .,\-]+$',::text)) NOT VALID;
```

```
--
```

```
-- TOC entry 3515 (class 2606 OID 16858)
```

```
-- Name: airline check_adress; Type: CHECK CONSTRAINT; Schema: schema;
Owner: postgres
```

```
--
```

```
ALTER TABLE schema.airline
```

```
    ADD CONSTRAINT check_adress CHECK (((address)::text ~ '^[A-Яa-яA-Za-
z0-9 .,\-/]+$',::text)) NOT VALID;
```

```
--
```

```
-- TOC entry 3555 (class 2606 OID 16917)
```

```
-- Name: seat_type check_base_price; Type: CHECK CONSTRAINT; Schema:
schema; Owner: postgres
```

```
--
```

```
ALTER TABLE schema.seat_type
```

```
    ADD CONSTRAINT check_base_price CHECK ((base_price > (0)::numeric))
NOT VALID;
```

```
--
```

```
-- TOC entry 3546 (class 2606 OID 16885)
```

```
-- Name: passenger check_birth_date; Type: CHECK CONSTRAINT; Schema:
schema; Owner: postgres
```

```
--
```

```
ALTER TABLE schema.passenger
```

```
    ADD CONSTRAINT check_birth_date CHECK ((birth_date <=
CURRENT_DATE)) NOT VALID;
```

```
--
```

```
-- TOC entry 3531 (class 2606 OID 16860)
```

```
-- Name: airport check_code_name; Type: CHECK CONSTRAINT; Schema:
schema; Owner: postgres
```

```
--
```

```
ALTER TABLE schema.airport
```

```
    ADD CONSTRAINT check_code_name CHECK (((code_name)::text ~ '^[A-
Z]{3,3}'::text)) NOT VALID;
```

```
--
```

```
-- TOC entry 3532 (class 2606 OID 16863)
```

```
-- Name: airport check_company_owner; Type: CHECK CONSTRAINT; Schema:
schema; Owner: postgres
```

```
--
```

```
ALTER TABLE schema.airport
```

```
    ADD CONSTRAINT check_company_owner CHECK
(((company_owner)::text ~ '^[A-Яа-яA-Za-z .,\-]+$'::text)) NOT VALID;
```

```
--
```

```
-- TOC entry 3533 (class 2606 OID 16862)
```

```
-- Name: airport check_country; Type: CHECK CONSTRAINT; Schema: schema;
Owner: postgres
```

```
--
```

```
ALTER TABLE schema.airport
```

```
    ADD CONSTRAINT check_country CHECK (((country)::text ~ '^[A-Яа-яA-
Za-z .,\-]+$'::text)) NOT VALID;
```

```
--
```

```
-- TOC entry 3548 (class 2606 OID 16887)
-- Name: passenger_passport check_country; Type: CHECK CONSTRAINT;
Schema: schema; Owner: postgres
--
```

```
ALTER TABLE schema.passenger_passport
  ADD CONSTRAINT check_country CHECK (((country)::text ~ '^[A-Яa-яA-Za-z .\,-]+$'::text)) NOT VALID;
```

```
--
-- TOC entry 3505 (class 2606 OID 16902)
-- Name: plane check_country; Type: CHECK CONSTRAINT; Schema: schema;
Owner: postgres
--
```

```
ALTER TABLE schema.plane
  ADD CONSTRAINT check_country CHECK (((country)::text ~ '^[A-Яa-яA-Za-z .\,-]+$'::text)) NOT VALID;
```

```
--
-- TOC entry 3516 (class 2606 OID 16856)
-- Name: airline check_country_reg; Type: CHECK CONSTRAINT; Schema:
schema; Owner: postgres
--
```

```
ALTER TABLE schema.airline
  ADD CONSTRAINT check_country_reg CHECK (((country_reg)::text ~ '^[A-Яa-яA-Za-z .\,-]+$'::text)) NOT VALID;
```

```
--
-- TOC entry 3556 (class 2606 OID 16916)
-- Name: seat_price_history check_dates; Type: CHECK CONSTRAINT; Schema:
schema; Owner: postgres
--
```

```
ALTER TABLE schema.seat_price_history
```

```
ADD CONSTRAINT check_dates CHECK ((start_date < end_date)) NOT  
VALID;
```

```
--  
-- TOC entry 3565 (class 2606 OID 16921)  
-- Name: service_price_history check_dates; Type: CHECK CONSTRAINT;  
Schema: schema; Owner: postgres  
--
```

```
ALTER TABLE schema.service_price_history  
ADD CONSTRAINT check_dates CHECK ((start_date < end_date)) NOT  
VALID;
```

```
--  
-- TOC entry 3541 (class 2606 OID 16925)  
-- Name: transit_stop check_dates; Type: CHECK CONSTRAINT; Schema:  
schema; Owner: postgres  
--
```

```
ALTER TABLE schema.transit_stop  
ADD CONSTRAINT check_dates CHECK ((departure_time < arrival_time))  
NOT VALID;
```

```
--  
-- TOC entry 3563 (class 2606 OID 16919)  
-- Name: service check_description; Type: CHECK CONSTRAINT; Schema:  
schema; Owner: postgres  
--
```

```
ALTER TABLE schema.service  
ADD CONSTRAINT check_description CHECK (((description)::text ~ '^[A-  
Яа-яA-Za-z .,\-]+$',::text)) NOT VALID;
```

```
--  
-- TOC entry 3535 (class 2606 OID 16911)
```

-- Name: route check_distance; Type: CHECK CONSTRAINT; Schema: schema;
Owner: postgres

--

ALTER TABLE schema.route

ADD CONSTRAINT check_distance CHECK ((distance > 0)) NOT VALID;

--

-- TOC entry 3517 (class 2606 OID 16846)

-- Name: airline check_email; Type: CHECK CONSTRAINT; Schema: schema;
Owner: postgres

--

ALTER TABLE schema.airline

ADD CONSTRAINT check_email CHECK (((email)::text ~ '^[A-Za-z0-9]+@[A-Za-z]+\.[a-z]{2,}\$'::text)) NOT VALID;

--

-- TOC entry 3506 (class 2606 OID 16908)

-- Name: plane check_flight_hours; Type: CHECK CONSTRAINT; Schema:
schema; Owner: postgres

--

ALTER TABLE schema.plane

ADD CONSTRAINT check_flight_hours CHECK ((flight_hours >= 0)) NOT
VALID;

--

-- TOC entry 3507 (class 2606 OID 16909)

-- Name: plane check_fuel_consumption; Type: CHECK CONSTRAINT; Schema:
schema; Owner: postgres

--

ALTER TABLE schema.plane

```
ADD CONSTRAINT check_fuel_consumption CHECK ((fuel_consumption > 0)) NOT VALID;
```

```
--
```

```
-- TOC entry 3520 (class 2606 OID 16866)
```

```
-- Name: employee check_full_name; Type: CHECK CONSTRAINT; Schema: schema; Owner: postgres
```

```
--
```

```
ALTER TABLE schema.employee
```

```
ADD CONSTRAINT check_full_name CHECK (((full_name)::text ~ '^[A-Яa-яA-Za-z .,\-]+$'::text)) NOT VALID;
```

```
--
```

```
-- TOC entry 3547 (class 2606 OID 16882)
```

```
-- Name: passenger check_full_name; Type: CHECK CONSTRAINT; Schema: schema; Owner: postgres
```

```
--
```

```
ALTER TABLE schema.passenger
```

```
ADD CONSTRAINT check_full_name CHECK (((full_name)::text ~ '^[A-Яa-яA-Za-z .,\-]+$'::text)) NOT VALID;
```

```
--
```

```
-- TOC entry 3523 (class 2606 OID 16872)
```

```
-- Name: employee_passport check_issued_address; Type: CHECK CONSTRAINT; Schema: schema; Owner: postgres
```

```
--
```

```
ALTER TABLE schema.employee_passport
```

```
ADD CONSTRAINT check_issued_address CHECK (((issued_address)::text ~ '^[A-Яa-яA-Za-z .,\-]+$'::text)) NOT VALID;
```

```

--
-- TOC entry 3549 (class 2606 OID 16889)
-- Name: passenger_passport check_issued_address; Type: CHECK
CONSTRAINT; Schema: schema; Owner: postgres
--

ALTER TABLE schema.passenger_passport
    ADD CONSTRAINT check_issued_address CHECK (((issued_address)::text ~
'^[A-Яa-яA-Za-z .,\-]+$::text)) NOT VALID;

--
-- TOC entry 3524 (class 2606 OID 16871)
-- Name: employee_passport check_issued_by; Type: CHECK CONSTRAINT;
Schema: schema; Owner: postgres
--

ALTER TABLE schema.employee_passport
    ADD CONSTRAINT check_issued_by CHECK (((issued_by)::text ~ '[A-Яa-
яA-Za-z .,\-]+$::text)) NOT VALID;

--
-- TOC entry 3550 (class 2606 OID 16888)
-- Name: passenger_passport check_issued_by; Type: CHECK CONSTRAINT;
Schema: schema; Owner: postgres
--

ALTER TABLE schema.passenger_passport
    ADD CONSTRAINT check_issued_by CHECK (((issued_by)::text ~ '[A-Яa-
яA-Za-z .,\-]+$::text)) NOT VALID;

--
-- TOC entry 3525 (class 2606 OID 16891)
-- Name: employee_passport check_issued_date; Type: CHECK CONSTRAINT;
Schema: schema; Owner: postgres
--

```

```
ALTER TABLE schema.employee_passport
  ADD CONSTRAINT check_issued_date CHECK ((issued_date <=
CURRENT_DATE)) NOT VALID;
```

```
--
-- TOC entry 3551 (class 2606 OID 16890)
-- Name: passenger_passport check_issued_date; Type: CHECK CONSTRAINT;
Schema: schema; Owner: postgres
--
```

```
ALTER TABLE schema.passenger_passport
  ADD CONSTRAINT check_issued_date CHECK ((issued_date <=
CURRENT_DATE)) NOT VALID;
```

```
--
-- TOC entry 3508 (class 2606 OID 16907)
-- Name: plane check_last_repair_date; Type: CHECK CONSTRAINT; Schema:
schema; Owner: postgres
--
```

```
ALTER TABLE schema.plane
  ADD CONSTRAINT check_last_repair_date CHECK ((last_repair_date <=
CURRENT_DATE)) NOT VALID;
```

```
--
-- TOC entry 3509 (class 2606 OID 16903)
-- Name: plane check_lifting_capacity; Type: CHECK CONSTRAINT; Schema:
schema; Owner: postgres
--
```

```
ALTER TABLE schema.plane
  ADD CONSTRAINT check_lifting_capacity CHECK ((lifting_capacity > 0))
NOT VALID;
```

```

--
-- TOC entry 3528 (class 2606 OID 16877)
-- Name: flight_crew check_medical_status; Type: CHECK CONSTRAINT;
Schema: schema; Owner: postgres
--

ALTER TABLE schema.flight_crew
    ADD CONSTRAINT check_medical_status CHECK (((medical_status)::text ~
'^[A-Яа-яA-Za-z .,\-]+$::text)) NOT VALID;

--
-- TOC entry 3518 (class 2606 OID 16852)
-- Name: airline check_name; Type: CHECK CONSTRAINT; Schema: schema;
Owner: postgres
--

ALTER TABLE schema.airline
    ADD CONSTRAINT check_name CHECK (((name)::text ~ '[A-Яа-яA-Za-z
.\-]+$::text)) NOT VALID;

--
-- TOC entry 3534 (class 2606 OID 16859)
-- Name: airport check_name; Type: CHECK CONSTRAINT; Schema: schema;
Owner: postgres
--

ALTER TABLE schema.airport
    ADD CONSTRAINT check_name CHECK (((name)::text ~ '[A-Яа-яA-Za-z
.\-]+$::text)) NOT VALID;

--
-- TOC entry 3564 (class 2606 OID 16918)

-- Name: service check_name; Type: CHECK CONSTRAINT; Schema: schema;
Owner: postgres
--

```

```
ALTER TABLE schema.service
  ADD CONSTRAINT check_name CHECK (((name)::text ~ '^[A-Яa-яA-Za-z
.,\-]+$',::text)) NOT VALID;
```

```
--
-- TOC entry 3526 (class 2606 OID 16873)
-- Name: employee_passport check_number_seria; Type: CHECK
CONSTRAINT; Schema: schema; Owner: postgres
--
```

```
ALTER TABLE schema.employee_passport
  ADD CONSTRAINT check_number_seria CHECK (((number_seria)::text ~
'^[0-9]{6,6}+[0-9]{4,4}$',::text)) NOT VALID;
```

```
--
-- TOC entry 3552 (class 2606 OID 16892)
-- Name: passenger_passport check_number_seria; Type: CHECK
CONSTRAINT; Schema: schema; Owner: postgres
--
```

```
ALTER TABLE schema.passenger_passport
  ADD CONSTRAINT check_number_seria CHECK (((number_seria)::text ~
'^[0-9]{6,6}+[0-9]{4,4}$',::text)) NOT VALID;
```

```
--
-- TOC entry 3558 (class 2606 OID 16924)
-- Name: ticket check_payment_date; Type: CHECK CONSTRAINT; Schema:
schema; Owner: postgres
--
```

```
ALTER TABLE schema.ticket
  ADD CONSTRAINT check_payment_date CHECK ((payment_date <=
CURRENT_DATE)) NOT VALID;
```

```
--
-- TOC entry 3559 (class 2606 OID 16922)
-- Name: ticket check_payment_status; Type: CHECK CONSTRAINT; Schema:
schema; Owner: postgres
--
```

```
ALTER TABLE schema.ticket
    ADD CONSTRAINT check_payment_status CHECK (((payment_status)::text ~
'^[A-Яa-яA-Za-z .\,-]+$'::text)) NOT VALID;
```

```
--
-- TOC entry 3510 (class 2606 OID 16904)
-- Name: plane check_performance; Type: CHECK CONSTRAINT; Schema:
schema; Owner: postgres
--
```

```
ALTER TABLE schema.plane
    ADD CONSTRAINT check_performance CHECK ((performance > 0)) NOT
VALID;
```

```
--
-- TOC entry 3519 (class 2606 OID 16845)
-- Name: airline check_phone_number; Type: CHECK CONSTRAINT; Schema:
schema; Owner: postgres
--
```

```
ALTER TABLE schema.airline
    ADD CONSTRAINT check_phone_number CHECK (((phone_number)::text
~~ '+'::text) AND ((phone_number)::text !~~ '+'::text) AND
((phone_number)::text ~ '^[0-9+]{12,14}$'::text))) NOT VALID;
```

```
--
-- TOC entry 3536 (class 2606 OID 16912)
-- Name: route check_plane_type; Type: CHECK CONSTRAINT; Schema:
schema; Owner: postgres
--
```

```
ALTER TABLE schema.route
    ADD CONSTRAINT check_plane_type CHECK (((plane_type)::text ~ '^[A-Яa-
яA-Za-z .,\-]+$'::text)) NOT VALID;
```

```
--
-- TOC entry 3539 (class 2606 OID 16875)
-- Name: flight check_planned_dates; Type: CHECK CONSTRAINT; Schema:
schema; Owner: postgres
--
```

```
ALTER TABLE schema.flight
    ADD CONSTRAINT check_planned_dates CHECK ((planned_departure <
planned_arrival)) NOT VALID;
```

```
--
-- TOC entry 3544 (class 2606 OID 16880)
-- Name: flight_transit_stop check_planned_dates; Type: CHECK CONSTRAINT;
Schema: schema; Owner: postgres
--
```

```
ALTER TABLE schema.flight_transit_stop
    ADD CONSTRAINT check_planned_dates CHECK ((planned_departure <
planned_arrival)) NOT VALID;
```

```
--
-- TOC entry 3521 (class 2606 OID 16867)
-- Name: employee check_position; Type: CHECK CONSTRAINT; Schema:
schema; Owner: postgres
--
```

```
ALTER TABLE schema.employee
    ADD CONSTRAINT check_position CHECK (((position)::text ~ '^[A-Яa-яA-
Za-z .,\-]+$'::text)) NOT VALID;
```

```
--
```

```
-- TOC entry 3557 (class 2606 OID 16915)
-- Name: seat_price_history check_price; Type: CHECK CONSTRAINT; Schema:
schema; Owner: postgres
```

```
--
```

```
ALTER TABLE schema.seat_price_history
    ADD CONSTRAINT check_price CHECK ((price >= (0)::numeric)) NOT
VALID;
```

```
--
```

```
-- TOC entry 3566 (class 2606 OID 16920)
-- Name: service_price_history check_price; Type: CHECK CONSTRAINT;
Schema: schema; Owner: postgres
```

```
--
```

```
ALTER TABLE schema.service_price_history
    ADD CONSTRAINT check_price CHECK ((price >= (0)::numeric)) NOT
VALID;
```

```
--
```

```
-- TOC entry 3511 (class 2606 OID 16905)
-- Name: plane check_purpose; Type: CHECK CONSTRAINT; Schema: schema;
Owner: postgres
```

```
--
```

```
ALTER TABLE schema.plane
    ADD CONSTRAINT check_purpose CHECK (((purpose)::text ~ '^[A-Яa-яA-
Za-z .,\-]+$',::text)) NOT VALID;
```

```
--
```

```
-- TOC entry 3529 (class 2606 OID 16878)
-- Name: flight_crew check_rejection_reason; Type: CHECK CONSTRAINT;
Schema: schema; Owner: postgres
```

```
--
```

```
ALTER TABLE schema.flight_crew
```

```
ADD CONSTRAINT check_rejection_reason CHECK (((rejection_reason)::text ~ '^[A-Яa-яA-Za-z .,\-%]+$'::text)) NOT VALID;
```

```
--  
-- TOC entry 3512 (class 2606 OID 16901)  
-- Name: plane check_seat_count; Type: CHECK CONSTRAINT; Schema:  
schema; Owner: postgres  
--
```

```
ALTER TABLE schema.plane  
ADD CONSTRAINT check_seat_count CHECK ((seat_count >= 0)) NOT  
VALID;
```

```
--  
-- TOC entry 3513 (class 2606 OID 16906)  
-- Name: plane check_speed; Type: CHECK CONSTRAINT; Schema: schema;  
Owner: postgres  
--
```

```
ALTER TABLE schema.plane  
ADD CONSTRAINT check_speed CHECK ((speed > 0)) NOT VALID;
```

```
--  
-- TOC entry 3562 (class 2606 OID 16864)  
-- Name: cashbox check_status; Type: CHECK CONSTRAINT; Schema: schema;  
Owner: postgres  
--
```

```
ALTER TABLE schema.cashbox  
ADD CONSTRAINT check_status CHECK (((status)::text ~ '^[A-Яa-яA-Za-z  
.,\-%]+$'::text)) NOT VALID;
```

```
--  
-- TOC entry 3540 (class 2606 OID 16874)
```

```
-- Name: flight check_status; Type: CHECK CONSTRAINT; Schema: schema;  
Owner: postgres
```

```
--
```

```
ALTER TABLE schema.flight
```

```
    ADD CONSTRAINT check_status CHECK (((status)::text ~ '^[A-Яa-яA-Za-z  
.,\-]+$':text)) NOT VALID;
```

```
--
```

```
-- TOC entry 3554 (class 2606 OID 16914)
```

```
-- Name: seat check_status; Type: CHECK CONSTRAINT; Schema: schema;  
Owner: postgres
```

```
--
```

```
ALTER TABLE schema.seat
```

```
    ADD CONSTRAINT check_status CHECK (((status)::text ~ '^[A-Яa-яA-Za-z  
.,\-]+$':text)) NOT VALID;
```

```
--
```

```
-- TOC entry 3542 (class 2606 OID 16926)
```

```
-- Name: transit_stop check_stop_time; Type: CHECK CONSTRAINT; Schema:  
schema; Owner: postgres
```

```
--
```

```
ALTER TABLE schema.transit_stop
```

```
    ADD CONSTRAINT check_stop_time CHECK ((stop_time > 0)) NOT  
VALID;
```

```
--
```

```
-- TOC entry 3537 (class 2606 OID 16913)
```

```
-- Name: route check_times; Type: CHECK CONSTRAINT; Schema: schema;  
Owner: postgres
```

```
--
```

```
ALTER TABLE schema.route
```

```
    ADD CONSTRAINT check_times CHECK ((departure_time < arrival_time))  
NOT VALID;
```

```
--  
-- TOC entry 3560 (class 2606 OID 16923)  
-- Name: ticket check_total_price; Type: CHECK CONSTRAINT; Schema:  
schema; Owner: postgres  
--
```

```
ALTER TABLE schema.ticket
```

```
    ADD CONSTRAINT check_total_price CHECK ((total_price > (0)::numeric))  
NOT VALID;
```

```
--  
-- TOC entry 3527 (class 2606 OID 16869)  
-- Name: employee_passport check_type; Type: CHECK CONSTRAINT; Schema:  
schema; Owner: postgres  
--
```

```
ALTER TABLE schema.employee_passport  
    ADD CONSTRAINT check_type CHECK (((type)::text ~ '^[A-Яa-яA-Za-z .\,-  
]+$'::text)) NOT VALID;
```

```
--  
-- TOC entry 3553 (class 2606 OID 16886)  
-- Name: passenger_passport check_type; Type: CHECK CONSTRAINT;  
Schema: schema; Owner: postgres  
--
```

```
ALTER TABLE schema.passenger_passport  
    ADD CONSTRAINT check_type CHECK (((type)::text ~ '^[A-Яa-яA-Za-z .\,-  
]+$'::text)) NOT VALID;
```

```
--  
-- TOC entry 3514 (class 2606 OID 16900)
```

-- Name: plane check_type; Type: CHECK CONSTRAINT; Schema: schema;
Owner: postgres

--

ALTER TABLE schema.plane

ADD CONSTRAINT check_type CHECK (((type)::text ~ '^[A-Яa-яA-Za-z .\,-
]+\$'::text)) NOT VALID;

--

-- TOC entry 3570 (class 2606 OID 16564)

-- Name: airline id_airline; Type: CONSTRAINT; Schema: schema; Owner:
postgres

--

ALTER TABLE ONLY schema.airline

ADD CONSTRAINT id_airline PRIMARY KEY (id_airline);

--

-- TOC entry 3600 (class 2606 OID 16766)

-- Name: cashbox id_cashbox; Type: CONSTRAINT; Schema: schema; Owner:
postgres

--

ALTER TABLE ONLY schema.cashbox

ADD CONSTRAINT id_cashbox PRIMARY KEY (id_cashbox);

--

-- TOC entry 3572 (class 2606 OID 16574)

-- Name: employee id_employee; Type: CONSTRAINT; Schema: schema; Owner:
postgres

--

ALTER TABLE ONLY schema.employee

ADD CONSTRAINT id_employee PRIMARY KEY (id_employee);

```
--  
-- TOC entry 3574 (class 2606 OID 16584)  
-- Name: employee_passport id_employee_passport; Type: CONSTRAINT;  
Schema: schema; Owner: postgres  
--
```

```
ALTER TABLE ONLY schema.employee_passport  
    ADD CONSTRAINT id_employee_passport PRIMARY KEY  
(id_employee_passport);
```

```
--  
-- TOC entry 3582 (class 2606 OID 16644)  
-- Name: flight id_flight; Type: CONSTRAINT; Schema: schema; Owner:  
postgres
```

```
--  
  
ALTER TABLE ONLY schema.flight  
    ADD CONSTRAINT id_flight PRIMARY KEY (id_flight);
```

```
--  
-- TOC entry 3576 (class 2606 OID 16594)  
-- Name: flight_crew id_flight_crew; Type: CONSTRAINT; Schema: schema;  
Owner: postgres  
--
```

```
ALTER TABLE ONLY schema.flight_crew  
    ADD CONSTRAINT id_flight_crew PRIMARY KEY (id_flight_crew);
```

```
--  
-- TOC entry 3586 (class 2606 OID 16684)  
-- Name: flight_transit_stop id_flight_transit_stop; Type: CONSTRAINT;  
Schema: schema; Owner: postgres  
--
```

```
ALTER TABLE ONLY schema.flight_transit_stop
  ADD CONSTRAINT id_flight_transit_stop PRIMARY KEY
(id_flight_transit_stop);
```

```
--
-- TOC entry 3590 (class 2606 OID 16709)
-- Name: passenger_passport id_passenger_passport; Type: CONSTRAINT;
Schema: schema; Owner: postgres
--
```

```
ALTER TABLE ONLY schema.passenger_passport
  ADD CONSTRAINT id_passenger_passport PRIMARY KEY
(id_passenger_passport);
```

```
--
-- TOC entry 3580 (class 2606 OID 16624)
```

```
-- Name: route id_route; Type: CONSTRAINT; Schema: schema; Owner: postgres
--
```

```
ALTER TABLE ONLY schema.route
  ADD CONSTRAINT id_route PRIMARY KEY (id_route);
```

```
--
-- TOC entry 3592 (class 2606 OID 16741)
-- Name: seat id_seat; Type: CONSTRAINT; Schema: schema; Owner: postgres
--
```

```
ALTER TABLE ONLY schema.seat
  ADD CONSTRAINT id_seat PRIMARY KEY (id_seat);
```

```
--
-- TOC entry 3596 (class 2606 OID 16751)
-- Name: seat_price_history id_seat_record; Type: CONSTRAINT; Schema:
schema; Owner: postgres
```

--

```
ALTER TABLE ONLY schema.seat_price_history
  ADD CONSTRAINT id_seat_record PRIMARY KEY (id_seat_record);
```

--

```
-- TOC entry 3594 (class 2606 OID 16724)
-- Name: seat_type id_seat_type; Type: CONSTRAINT; Schema: schema; Owner:
postgres
```

--

```
ALTER TABLE ONLY schema.seat_type
  ADD CONSTRAINT id_seat_type PRIMARY KEY (id_seat_type);
```

--

```
-- TOC entry 3602 (class 2606 OID 16791)
-- Name: service id_service; Type: CONSTRAINT; Schema: schema; Owner:
postgres
```

--

```
ALTER TABLE ONLY schema.service
  ADD CONSTRAINT id_service PRIMARY KEY (id_service);
```

--

```
-- TOC entry 3606 (class 2606 OID 16814)
-- Name: service_price_history id_service_record; Type: CONSTRAINT; Schema:
schema; Owner: postgres
```

--

```
ALTER TABLE ONLY schema.service_price_history
  ADD CONSTRAINT id_service_record PRIMARY KEY (id_service_record);
```

--

```
-- TOC entry 3598 (class 2606 OID 16761)
```

```
-- Name: ticket id_ticket; Type: CONSTRAINT; Schema: schema; Owner:
postgres
```

```
--
```

```
ALTER TABLE ONLY schema.ticket
  ADD CONSTRAINT id_ticket PRIMARY KEY (id_ticket);
```

```
--
```

```
-- TOC entry 3604 (class 2606 OID 16796)
```

```
-- Name: ticket_service id_ticket_service; Type: CONSTRAINT; Schema: schema;
Owner: postgres
```

```
--
```

```
ALTER TABLE ONLY schema.ticket_service
  ADD CONSTRAINT id_ticket_service PRIMARY KEY (id_ticket_service);
```

```
--
```

```
-- TOC entry 3584 (class 2606 OID 16669)
```

```
-- Name: transit_stop id_transit_stop; Type: CONSTRAINT; Schema: schema;
Owner: postgres
```

```
--
```

```
ALTER TABLE ONLY schema.transit_stop
```

```
  ADD CONSTRAINT id_transit_stop PRIMARY KEY (id_transit_stop);
```

```
--
```

```
-- TOC entry 3588 (class 2606 OID 16704)
```

```
-- Name: passenger passenger_pkey; Type: CONSTRAINT; Schema: schema;
Owner: postgres
```

```
--
```

```
ALTER TABLE ONLY schema.passenger
  ADD CONSTRAINT passenger_pkey PRIMARY KEY (id_passenger);
```

```
--  
-- TOC entry 3568 (class 2606 OID 16559)  
-- Name: plane plane_pkey; Type: CONSTRAINT; Schema: schema; Owner:  
postgres  
--
```

```
ALTER TABLE ONLY schema.plane  
    ADD CONSTRAINT plane_pkey PRIMARY KEY (id_plane);
```

```
--  
-- TOC entry 3608 (class 2606 OID 16575)  
-- Name: employee id_airline; Type: FK CONSTRAINT; Schema: schema;  
Owner: postgres  
--
```

```
ALTER TABLE ONLY schema.employee  
    ADD CONSTRAINT id_airline FOREIGN KEY (id_airline) REFERENCES  
schema.airline(id_airline);
```

```
--  
-- TOC entry 3607 (class 2606 OID 16565)  
-- Name: plane id_airline; Type: FK CONSTRAINT; Schema: schema; Owner:  
postgres  
--
```

```
ALTER TABLE ONLY schema.plane
```

```
    ADD CONSTRAINT id_airline FOREIGN KEY (id_airline) REFERENCES  
schema.airline(id_airline) NOT VALID;
```

```
--  
-- TOC entry 3612 (class 2606 OID 16625)  
-- Name: route id_airline; Type: FK CONSTRAINT; Schema: schema; Owner:  
postgres  
--
```

```
ALTER TABLE ONLY schema.route
  ADD CONSTRAINT id_airline FOREIGN KEY (id_airline) REFERENCES
schema.airline(id_airline);
```

```
--
-- TOC entry 3618 (class 2606 OID 16675)
-- Name: transit_stop id_airport; Type: FK CONSTRAINT; Schema: schema;
Owner: postgres
--
```

```
ALTER TABLE ONLY schema.transit_stop
  ADD CONSTRAINT id_airport FOREIGN KEY (id_airport) REFERENCES
schema.airport(id_airport);
```

```
--
-- TOC entry 3614 (class 2606 OID 16635)
-- Name: route id_arrival_airport; Type: FK CONSTRAINT; Schema: schema;
Owner: postgres
--
```

```
ALTER TABLE ONLY schema.route
  ADD CONSTRAINT id_arrival_airport FOREIGN KEY (id_arrival_airport)
REFERENCES schema.airport(id_airport);
```

```
--
-- TOC entry 3627 (class 2606 OID 16767)
-- Name: ticket id_cashbox; Type: FK CONSTRAINT; Schema: schema; Owner:
postgres
--
```

```
ALTER TABLE ONLY schema.ticket
  ADD CONSTRAINT id_cashbox FOREIGN KEY (id_cashbox)
REFERENCES schema.cashbox(id_cashbox) NOT VALID;
```

```
--
-- TOC entry 3613 (class 2606 OID 16630)
-- Name: route id_departure_airport; Type: FK CONSTRAINT; Schema: schema;
Owner: postgres
--
```

```
ALTER TABLE ONLY schema.route
  ADD CONSTRAINT id_departure_airport FOREIGN KEY
(id_departure_airport) REFERENCES schema.airport(id_airport);
```

```
--
-- TOC entry 3609 (class 2606 OID 16585)
-- Name: employee_passport id_employee; Type: FK CONSTRAINT; Schema:
schema; Owner: postgres
--
```

```
ALTER TABLE ONLY schema.employee_passport
  ADD CONSTRAINT id_employee FOREIGN KEY (id_employee)
REFERENCES schema.employee(id_employee);
```

```
--
-- TOC entry 3611 (class 2606 OID 16660)
-- Name: flight_crew id_employee_passport; Type: FK CONSTRAINT; Schema:
schema; Owner: postgres
--
```

```
ALTER TABLE ONLY schema.flight_crew
  ADD CONSTRAINT id_employee_passport FOREIGN KEY
(id_employee_passport) REFERENCES
schema.employee_passport(id_employee_passport) NOT VALID;
```

```
--
-- TOC entry 3610 (class 2606 OID 16655)
-- Name: flight_crew id_flight; Type: FK CONSTRAINT; Schema: schema;
Owner: postgres
```

--

```
ALTER TABLE ONLY schema.flight_crew
  ADD CONSTRAINT id_flight FOREIGN KEY (id_flight) REFERENCES
schema.flight(id_flight) NOT VALID;
```

--

```
-- TOC entry 3619 (class 2606 OID 16685)
-- Name: flight_transit_stop id_flight; Type: FK CONSTRAINT; Schema: schema;
Owner: postgres
```

--

```
ALTER TABLE ONLY schema.flight_transit_stop
  ADD CONSTRAINT id_flight FOREIGN KEY (id_flight) REFERENCES
schema.flight(id_flight);
```

--

```
-- TOC entry 3623 (class 2606 OID 16730)
-- Name: seat id_flight; Type: FK CONSTRAINT; Schema: schema; Owner:
postgres
```

--

```
ALTER TABLE ONLY schema.seat
  ADD CONSTRAINT id_flight FOREIGN KEY (id_flight) REFERENCES
schema.flight(id_flight) NOT VALID;
```

--

```
-- TOC entry 3622 (class 2606 OID 16710)
-- Name: passenger_passport id_passenger; Type: FK CONSTRAINT; Schema:
schema; Owner: postgres
```

--

```
ALTER TABLE ONLY schema.passenger_passport
```

```
ADD CONSTRAINT id_passenger FOREIGN KEY (id_passenger)
REFERENCES schema.passenger(id_passenger);
```

```
--
-- TOC entry 3628 (class 2606 OID 16772)
-- Name: ticket id_passenger_passport; Type: FK CONSTRAINT; Schema:
schema; Owner: postgres
--
```

```
ALTER TABLE ONLY schema.ticket
ADD CONSTRAINT id_passenger_passport FOREIGN KEY
(id_passenger_passport) REFERENCES
schema.passenger_passport(id_passenger_passport) NOT VALID;
```

```
--
-- TOC entry 3616 (class 2606 OID 16650)
-- Name: flight id_plane; Type: FK CONSTRAINT; Schema: schema; Owner:
postgres
--
```

```
ALTER TABLE ONLY schema.flight
ADD CONSTRAINT id_plane FOREIGN KEY (id_plane) REFERENCES
schema.plane(id_plane);
```

```
--
-- TOC entry 3615 (class 2606 OID 16645)
-- Name: flight id_route; Type: FK CONSTRAINT; Schema: schema; Owner:
postgres
--
```

```
ALTER TABLE ONLY schema.flight
ADD CONSTRAINT id_route FOREIGN KEY (id_route) REFERENCES
schema.route(id_route);
```

```
--
-- TOC entry 3621 (class 2606 OID 16695)
-- Name: flight_transit_stop id_route; Type: FK CONSTRAINT; Schema: schema;
Owner: postgres
```

--

```
ALTER TABLE ONLY schema.flight_transit_stop
    ADD CONSTRAINT id_route FOREIGN KEY (id_route) REFERENCES
schema.route(id_route);
```

--

```
-- TOC entry 3625 (class 2606 OID 16735)
-- Name: seat_type id_route; Type: FK CONSTRAINT; Schema: schema; Owner:
postgres
```

--

```
ALTER TABLE ONLY schema.seat_type
    ADD CONSTRAINT id_route FOREIGN KEY (id_route) REFERENCES
schema.route(id_route) NOT VALID;
```

--

```
-- TOC entry 3617 (class 2606 OID 16670)
-- Name: transit_stop id_route; Type: FK CONSTRAINT; Schema: schema;
Owner: postgres
```

--

```
ALTER TABLE ONLY schema.transit_stop
    ADD CONSTRAINT id_route FOREIGN KEY (id_route) REFERENCES
schema.route(id_route);
```

--

```
-- TOC entry 3629 (class 2606 OID 16777)
-- Name: ticket id_seat; Type: FK CONSTRAINT; Schema: schema; Owner:
postgres
```

--

```
ALTER TABLE ONLY schema.ticket
    ADD CONSTRAINT id_seat FOREIGN KEY (id_seat) REFERENCES
schema.seat(id_seat) NOT VALID;
```

--

-- TOC entry 3624 (class 2606 OID 16742)
-- Name: seat id_seat_type; Type: FK CONSTRAINT; Schema: schema; Owner:
postgres

--

ALTER TABLE ONLY schema.seat
ADD CONSTRAINT id_seat_type FOREIGN KEY (id_seat_type)
REFERENCES schema.seat_type(id_seat_type) NOT VALID;

--

-- TOC entry 3626 (class 2606 OID 16752)
-- Name: seat_price_history id_seat_type; Type: FK CONSTRAINT; Schema:
schema; Owner: postgres

--

ALTER TABLE ONLY schema.seat_price_history
ADD CONSTRAINT id_seat_type FOREIGN KEY (id_seat_type)
REFERENCES schema.seat_type(id_seat_type);

--

-- TOC entry 3630 (class 2606 OID 16782)
-- Name: ticket id_seat_type; Type: FK CONSTRAINT; Schema: schema; Owner:
postgres

--

ALTER TABLE ONLY schema.ticket
ADD CONSTRAINT id_seat_type FOREIGN KEY (id_seat_type)
REFERENCES schema.seat_type(id_seat_type) NOT VALID;

--

-- TOC entry 3633 (class 2606 OID 16815)

-- Name: service_price_history id_service; Type: FK CONSTRAINT; Schema: schema; Owner: postgres

--

ALTER TABLE ONLY schema.service_price_history

ADD CONSTRAINT id_service FOREIGN KEY (id_service) REFERENCES schema.service(id_service);

--

-- TOC entry 3631 (class 2606 OID 16797)

-- Name: ticket_service id_service; Type: FK CONSTRAINT; Schema: schema; Owner: postgres

--

ALTER TABLE ONLY schema.ticket_service

ADD CONSTRAINT id_service FOREIGN KEY (id_service) REFERENCES schema.service(id_service);

--

-- TOC entry 3632 (class 2606 OID 16802)

-- Name: ticket_service id_ticket; Type: FK CONSTRAINT; Schema: schema; Owner: postgres

--

ALTER TABLE ONLY schema.ticket_service

ADD CONSTRAINT id_ticket FOREIGN KEY (id_ticket) REFERENCES schema.ticket(id_ticket);

--

-- TOC entry 3620 (class 2606 OID 16690)

-- Name: flight_transit_stop id_transit_stop; Type: FK CONSTRAINT; Schema: schema; Owner: postgres

--

```
ALTER TABLE ONLY schema.flight_transit_stop
  ADD CONSTRAINT id_transit_stop FOREIGN KEY (id_transit_stop)
REFERENCES schema.transit_stop(id_transit_stop);
```

```
-- Completed on 2026-03-30 13:03:29 MSK
```

```
--
```

```
-- PostgreSQL database dump complete
```

```
--
```

```
\unrestrict
```

```
wM96tMIVekC04YxgAZzNHuX3LhcoxbsVtJ7S4wQFoFBZOvFPTQmOTYDq7  
CPa0W5
```

Выводы:

В ходе выполнения лабораторных работ были успешно освоены основные этапы проектирования и реализации баз данных

Изучены и применены на практике:

- преобразование инфологической модели в физическую структуру БД • замена составных ключей на суррогатные
- установка различных ограничений: PRIMARY KEY, FOREIGN KEY, NOT NULL, CHECK
- наполнение таблиц тестовыми данными • создание резервных копий (Custom и Plain) и восстановление из них.

Практически отработаны возможности pgAdmin 4: визуальное проектирование ERD, выполнение SQL-запросов, настройка параметров резервного копирования. Полученные навыки позволяют в дальнейшем разрабатывать полноценные реляционные базы данных для информационных систем различного назначения.